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Rare earth metals detection and recognition based on laser induced breakdown spectroscopy and machine learning

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Abstract: The rapid detection and identification of the electronic waste (e-waste) containing rare earth (RE) elements is of great significance for the recycling of RE elements. However, the analysis of these materials is extremely challenging due to extreme similarities in appearance or chemical composition. In this research, a new system based on laser induced breakdown spectroscopy (LIBS) and machine learning algorithms is developed for identifying and classifying e-waste of rare-earth phosphors (REPs). Three different kinds of phosphors are selected and the spectra is monitored using this new developed system. The analysis of phosphor spectra shows that there are Gd, Yd, and Y RE element spectra in the phosphor. The results also verify that LIBS could be used to detect RE elements. An unsupervised learning method, principal component analysis (PCA), is used to distinguish the three phosphors and training data set is stored for further identification. Additionally, a supervised learning method, backpropagation artificial neural network (BP-ANN) algorithm is used to establish a neural network model to identify phosphors. The result show that the final phosphor recognition rate reaches 99.9%. The innovative system based on LIBS and machine learning (ML) has the potential to improve rapid in situ detection of RE elements for the classification of e-waste.

1. Introduction

REPs, due to their noteworthy advantages including the high luminance efficiency, good chemical stability, and strong luminance brightness, etc., has immense marked demand in the application of solid-state lighting, computed tomography tubes, television screens, display panels, biomedicine, and forensics [1–5]. As a consequence of the rapid updating and iteration of electronic products, large quantities of electronic waste (e-waste) containing rare-earth (RE) metals, including REPs, are now among the most important electronic pollutants worldwide. Several studies have shown that long-term exposure to REP e-waste in the environment will cause environmental imbalance and can cause great harm to the human body, largely because it contains large numbers of REPs, oxides of REPs, and toxic metals [6]. Therefore, a method that can detect these phosphors and their RE elements quickly and accurately is urgently required.

At present, the most common techniques used for optical characterization of substance containing RE elements include photoluminescence spectroscopy (PL), Raman spectroscopy, and

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Fourier transform infrared spectroscopy (FTIR) [7]. These methods are mainly used to measure the vibration spectra and luminescence characteristics of REP materials. From the results, the properties of the REP materials can be determined and their potential use in many fields in industry and technology can be identified. PL spectroscopy is a nondestructive and noncontact optical method for probing of the electronic structures of materials and has been used to measure the spectral features of cerium and samarium when co-doped in yttrium–aluminum–garnet ceramics [8]. Raman spectroscopy, which is a type of scattering spectroscopy, is primarily used to study molecular structures. For example, Kolesnikov et al. presented the spectral of synthesized YVO^4:Dy^{3+}1 powder as measured by Raman spectroscopy [9]. FTIR spectroscopy is a technology that uses the absorption properties of infrared light to determine the structure and composition of molecules. For example, Zhang et al. reported the FTIR spectral of $La_3Ga_5SiO_{14}$:Sm^{3+} phosphors [10]. Although many detection methods are available for REPs, there are few techniques that can detect these spectra and classify them at the same time; such a technique would be highly significant for classification and recycling of e-waste containing RE elements.

Laser-induced breakdown spectroscopy (LIBS) is an elemental analysis technique based on atomic emission spectroscopy. LIBS offers outstanding advantages in multi-element analysis and enable real-time, rapid in situ measurements, which means that it has been investigated extensively and is used widely in fields such as deep-sea exploration, geological exploration, biomedicine and environmental monitoring [11–15]. Remarkably, the various spectral data measured by LIBS can usually be classified by after being processed by machine learning algorithms. For example, Ye et al. realized detection and classification of four different combustion substances using LIBS [16], while Chen et al. identified four different pencil marks, which is highly significant for forensics applications [17]. Qu et al. used the method to achieve cluster analysis and classification by extracting the features of various incenses [18]. Manard et al. quantified the RE elements Eu, Nd and Yb that were present in a uranium oxide matrix using a portable handheld laser-induced breakdown spectrometer (HH LIBS) [19]. Martin used LIBS to study the optical emission functions of RE elements, including europium (Eu), gadolinium (Gd), lanthanum (La), praseodymium (Pr), neodymium (Nd), and samarium (Sm) [20]. Unnikrishnan et al. reported a method to increase the detection limits for trace elements in liquids using LIBS technology [21]. Rao et al. introduced LIBS technology in combination with machine learning algorithms for characterization of nuclear materials [22]. Harmon introduced the application of LIBS technology to elemental detection, sample classification, and rock characterization, with results that indicated the potential of LIBS technology for use in mineral exploration applications [23]. Zhao reported a LIBS method for Pb detection and their results show that LIBS technology can be used to detect heavy metals in soil [24]. Bhatt reported a multivariate calibration strategy that combined machine-learning (ML) technology with LIBS, for the quantification of trace uranium in cellulose and uranium-bearing mineral ores [25]. In the past, LIBS has not been widely in identification and classification of e-waste, particularly for REPs, because of the high ionization energies of some elements, low laser energy, high relative noise, and other factors. However, with the development of ML, researchers have gradually used ML algorithms to achieve spectral classification by analyzing the abundant quantities of LIBS data based on train models. This approach realizes intelligence spectral detection and has provided new ideas for identification and classification of REPs.

In this research, a new system (ML-LIBS) for e-waste detection and classification is firstly established by combining LIBS with ML to determine the REP compositions and recognition rates. Consider three different phosphor types, designated CRT-B, CRT-G and IR-G (see Table 1), as an example. First, the spectral data of the different phosphor types are obtained using the self-designed ML-LIBS system. Then, the spectral data are obtained after wavelength drift calibration by comparing the original measured data with the characteristic spectral lines of the elements in the National Institute of Standards and Technology (NIST) database. Subsequently, by comparing the spectrograms of the three phosphor types, the featured spectral lines with large differences in types and intensity are selected and their dimensionality is reduced by applying a dimensionality reduction technique, PCA, which is an unsupervised data dimensionality reduction method. According to the scores and scatter distributions of the first three principal components, classification of the three REPs can be performed preliminarily and they can be distinguished qualitatively. Finally, based on the principal component data obtained via PCA, the total test set and respective recognition rates for the three REPs are obtained by using a backpropagation artificial neural network (BP-ANN) to classify them quantitatively.

Table 1. Characteristics of experimental samples

Type	Composition	Power color	Granularity
CRT-B	ZnS: Ag	white	3.0
CRT-G	GdOS: Tb	white	3.5
IR-G	NaYF: Yb: Tm	white	3.7

2. Experiment

2.1. Experimental setup

A schematic diagram of the experimental station is shown in Fig. 1. The station contains a laser generation system, a signal detection system and a spectral analysis system. The laser generation system comprises a nanosecond Q-switched Nd:YAG laser (Continuum Surelite II-10) and a series of lenses, while the signal detection system includes a four-channel spectrometer (AvaSpec-USL2048-4 Channel-usb2.0, Avantes), a multi-channel signal sequencer, and a 3D movable load stage. In the laser generation system, the nanosecond Q-switched Nd: YAG laser operates at a fundamental wavelength of 1064 nm with a repetition rate of 10 Hz, a full width at half maximum (FWHM) of 6 ns, maximum energy of 200 mJ, and a beam diameter of 5 mm. Following the laser, a series of lenses is set into the optical path after the target. Then, a highly focused laser pulse irradiates the pressed phosphor. The spectra of the targets are recorded by the spectrometer. In the signal detection system, the four-channel spectrometer operates at a spectral resolution of 0.07 nm with a spectral window ranging from 200 to 900 nm, and an integration time of 2 ns. To reduce the influence of random errors on the experiment results, 60 measurement results are used to obtain each spectrum for analysis. To control the signal acquisition time delay, a digital generator is set between the pulsed laser system and the spectrometer.

The LIBS system has been introduced previously and a great deal of research work into the LIBS detection technique has been completed. For example, Zhou et al. and Qu et al. used LIBS to monitor carbon concentrations and to detect and analyze local air pollution, respectively [26,27]. The LIBS system was also applied in combination with a self-designed single-particle aerosol mass spectrometry (SPAMS) system to direct online detection of VOCs in the atmosphere [28].

2.2. Sample preparation

When a high-quality phosphor with reasonable light radiance is sought, a wide variety of these materials is available in the world market. The selection includes phosphors developed by Harris Technology (USA), Phosphor Technology (UK), Nichia (Japan), and Shanghai Keyan Phosphor Technology (KPT), China. In view of the application of such phosphors in vacuum electrics worldwide, this paper studies the spectra of several phosphors that are commonly used as image enhancer phosphors. To verify the superiority of the proposed system in terms of its spectrum detection and substance classification performance, three widely used phosphor types, comprising

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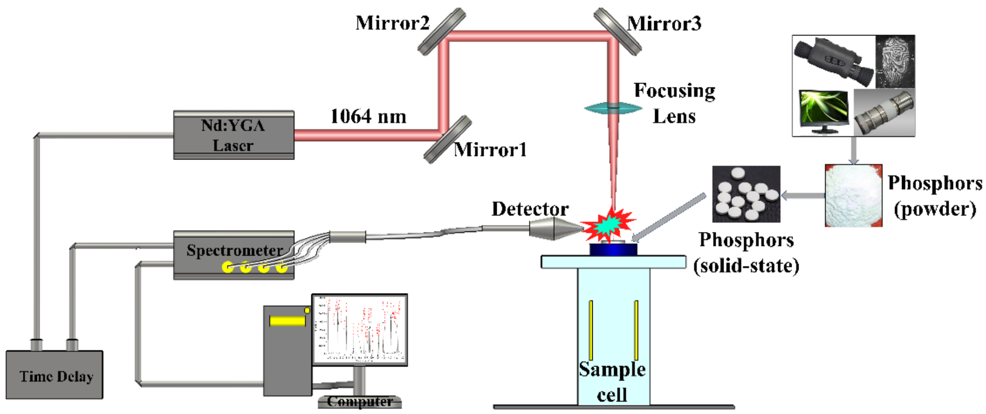

Fig. 1. Schematic diagram of the experiment system.

CRT-B, CRT-G, and IR-B, were selected; all of these phosphors were purchased from KPT. Among the three materials, CRT-G and IR-G contain RE metal elements (i.e., they are REPs), while CRT-B does not contain RE elements. The main physical information for the three samples is shown in Table 1. Because of the high similarity among the physical properties of the three phosphors, it is difficult to distinguish them directly using the naked eye.

Because the phosphor samples used in the experiments are all solid-state powders, the phosphors must be processed into thin cylindrical-shaped solids before performing the LIBS experiment to avoid splashing during laser irradiation. The laminating method used in this study is one of the most commonly used phosphor processing methods at present. The specific sample preparation process is described as follows. The dried powder is introduced into a special die used for tablet pressing, pressed at up to 4 Pa on the tablet pressing machine, with this pressure being maintained for 2–3 min, and then a disc-shaped wafer with a thickness of approximately 3 mm and a diameter of approximately 12 mm is realized as the sample for the experiments.

3. Results and discussions

3.1. Rapid determination of phosphor elements

The spectrum of the e-waste REP from IR-G was measured in an air environment. Because the experiment was performed directly in air without any pretreatment, the spectra of elements with higher contents in ambient air, i.e., N, O, and H, will have a specific impact on the LIBS detection performance. Studies have shown that the strong spectral lines for the N, O, and H elements are mainly concentrated in a band above 500 nm. To reduce the influence of the elements in the air, the spectrum from 220–470 nm was mainly tested. With reference to the NIST data and related research, we calibrated the spectral drift characteristics and emission peaks caused by the ambient temperature of the spectrometer, the fiber materials, and out-of-synchronization delay times. The emission spectrum of the IR-G phosphor, which covers the 220–470 nm wavelength range, is shown in Fig. 2. Note that very rich RE metal lines for Gd, Y, and Yd are present in the IR-G spectrum. Among these lines, the spectral lines of Y are most numerous and have high intensity, thus indicating that the RE element Y is the main constituent element of the IR-G phosphor. Because of the superior properties of RE elements, they are usually used as activators for phosphors. In addition, a small number of spectral lines for Al, Fe, and Ca can be observed in Fig. 2(a), (b), (c), and (d), indicating that the corresponding elements are contained within the phosphor.

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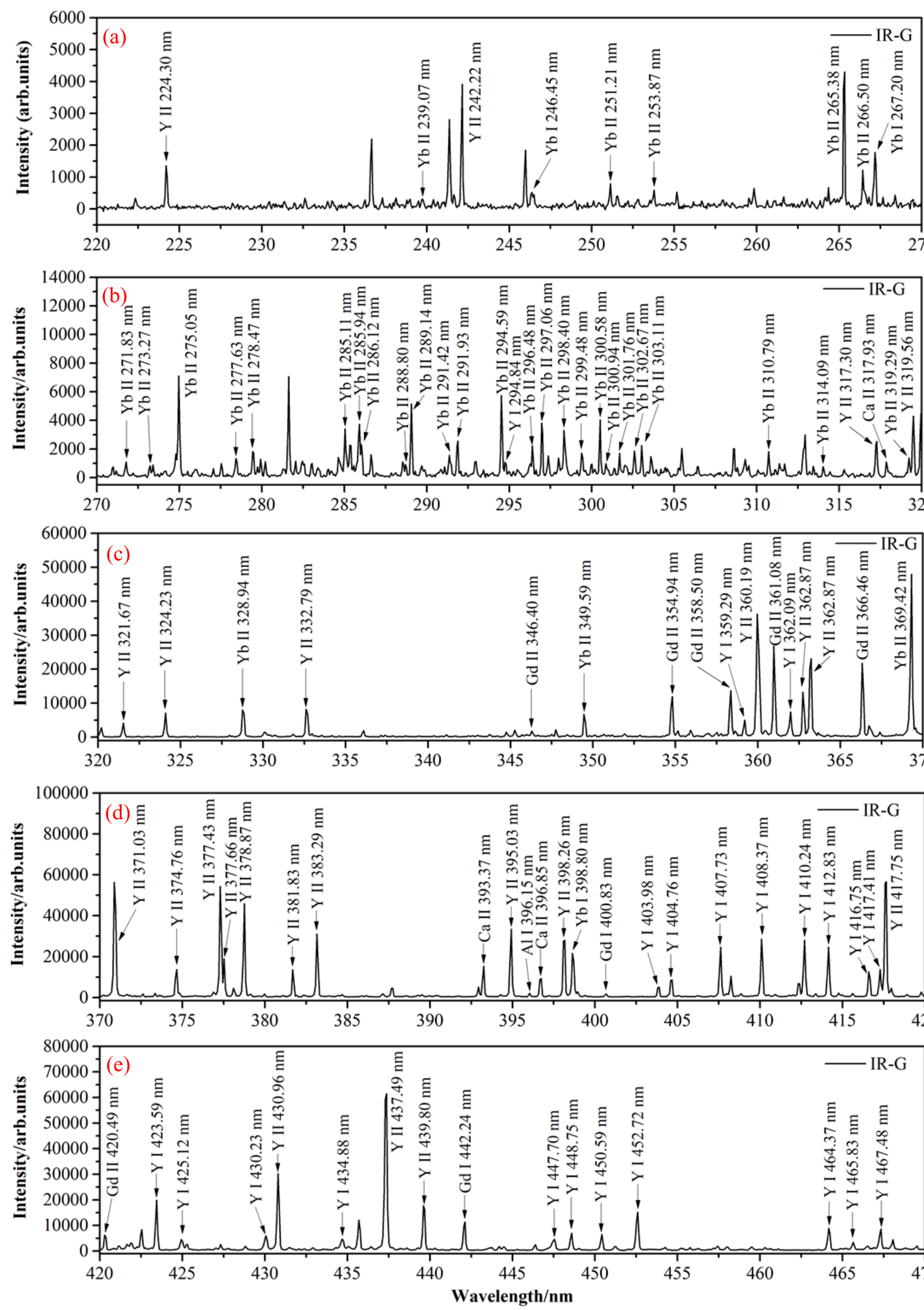

Fig. 2. The spectra of IR-G from 220–470 nm: (a)220–270 nm, (b) 270–320 nm, (c) 320–370 nm, (d) 370–420 nm, and (e) 420–470 nm.

Figure 3 shows the characteristic spectral lines for three different phosphors, including CRT-B, CRT-G, and IR-G, from 270 nm to 340 nm. Here, CRT-B is a metal-activated phosphor, while both CRT-G and IR-G are RE metal-activated phosphors. As shown in Fig. 3(a), the spectral lines of the metallic elements Fe, Zn, Mg, Al, and that of the metalloid element Si are detected in the

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CRT-B phosphor. Note here that Zn is a host, Ag is a dopant, and the others are microelements. In Fig. 3(b), the spectral lines of Gd are numerous and are high in intensity; at the same time, spectral lines for Fe are also detected, but they are both small in numbers and low in intensity. In Fig. 3(c), we observed large numbers of high-intensity spectral lines for the RE elements Yb and Y, along with a small number of Fe and Ca spectral lines. In comparison to the metal iondoped phosphors of CRT-B, the RE-doped phosphors of CRT-G and IR-G are quite advantageous of high emission efficiency as well as abundant spectral lines.

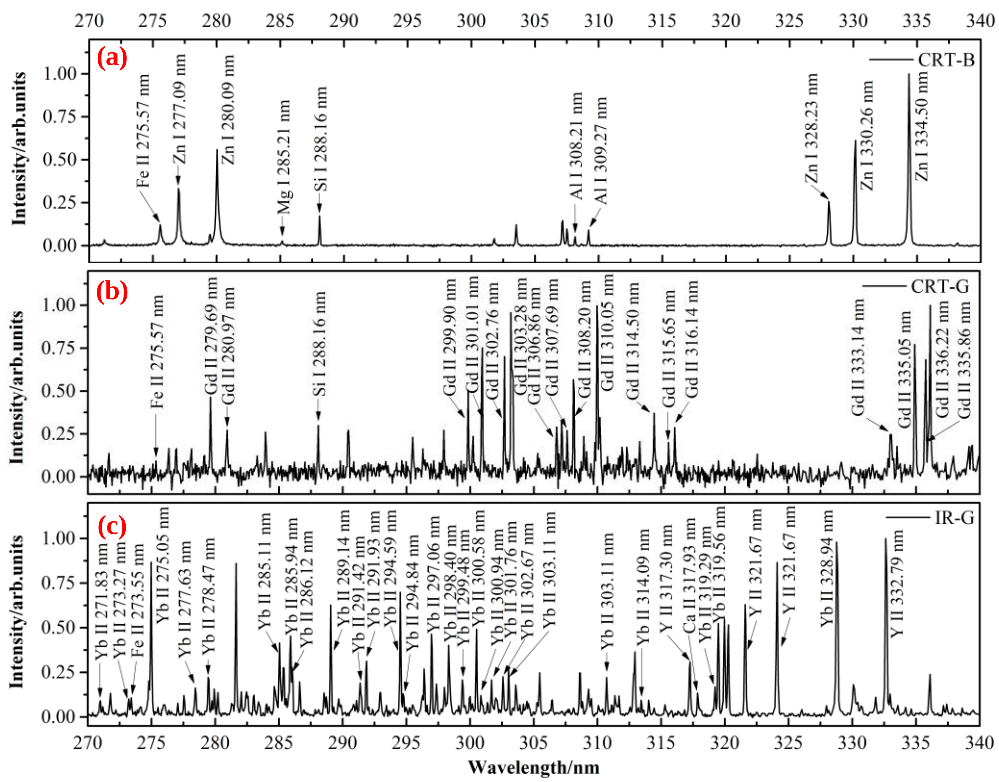

Fig. 3. Spectral for different kinds of phosphors from 270 nm to 340 nm: (a) CRT-B, (b) CRT-G, and (c) IR-B.

3.2. Phosphor recognition based on unsupervised ML

As an unsupervised learning method, PCA converts original variables into new orthogonal variables through a linear transformation procedure, which can be used for reduction and multivariate data analysis. In the experiments in this work, 2056 sets of spectral data for CRT-G, CRT-B, and IR-G were measured. The spectra contain unique information about each phosphor and are equivalent to the fingerprint of each phosphor. Therefore, classification of the phosphors can be achieved through deep mining and analysis of the spectral data. 100 groups were selected from the spectrum for each phosphor as sample data for PCA and BP-ANN model processing. There were thus 300 groups of data samples in total for the three phosphors.

Before further processing, the commonly used range normalization method was applied to standardize all the spectral intensity data and all the intensity values were converted into [0, 1]. The specific operating method is to find the maximum ($X_{\max}$) and minimum ($X_{\min}$) in the spectral intensity, calculate the range ($R = X_{\max} - X_{\min}$), subtract the minimum ($X_{\min}$) from each observed

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value (X), and then divide by the range (R). Equation (1) was used to normalize the data.

$$X' = \frac{(X - X_{\min})}{X_{\max} - X_{\min}} \tag{1}$$

In the experiments, based on consideration of the characteristics of the different targets, spectral lines of different types and intensities were selected as the characteristic parameters of the various phosphors. According to previous spectral calibrations, certain atomic or ion emission lines for Zn, Gd, and Y were selected for the analysis, as shown in Table 2. The peak intensity was selected as the original variable for the PCA, i.e., as the feature of the ML algorithm (because of the large numbers of spectral lines, only certain specific spectral lines are listed in the table). In this paper, unsupervised and supervised ML is performed based on these features of the observed values for the different spectral intensities.

Figure 4 shows the contribution rates and cumulative contribution rates of the nine principal components that were obtained based on the characteristic spectral lines given in Table 2. The contribution rates of the first three principal components were 59.9%, 31.8%, and 7.6%, respectively. In addition, the cumulative contribution rate of these components reached 99.3%, thus indicating that the first three principal components carried most of the information of the original variable.

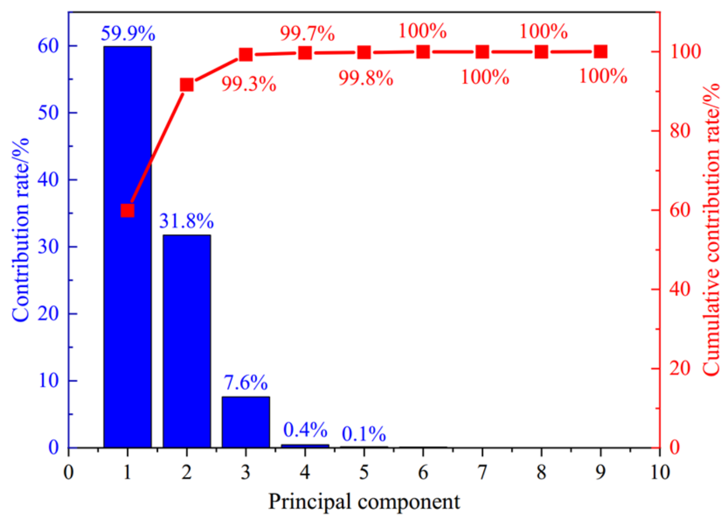

Fig. 4. Contributions and cumulative contribution rates of principal components.

Table 2. Characteristic wavelength

Element	Wavelength/nm
Zn	330.167, 334.361, 472.156
Gd	303.193, 376.716, 404.821
Y	370.877, 377.288, 437.369

Using the scores for the first three principal components, a 3D scatter distribution for the different phosphor samples can be formed, as shown in Fig. 5. As Fig. 5 clearly shows, the sample points for each of the phosphor aggregate clusters are separated from each other and can be distinguished directly. This indicates that the spectral intensities obtained in the experiments can be used to distinguish the three phosphors.

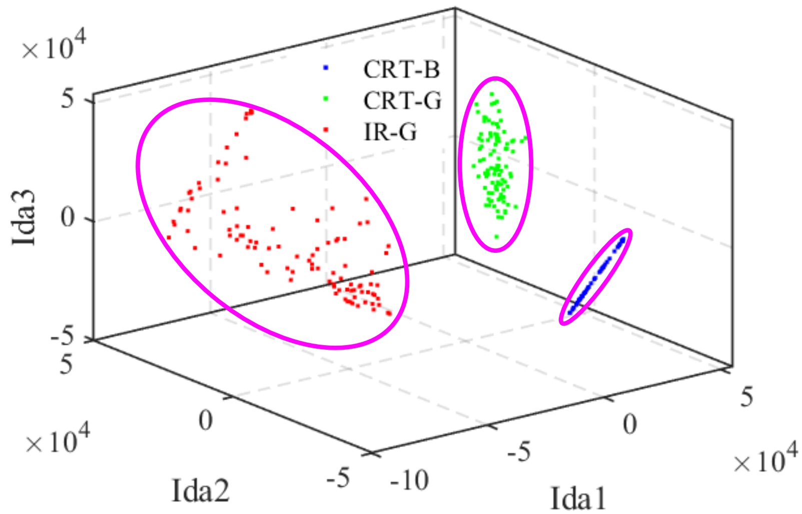

Fig. 5. Scatter distribution of different phosphor samples in score space of the first 3 PCs: The sample point covered by the blue square is basically the spectrum of CRT-B, the green triangular covers the spectrum of CRT-G, while the red pentagram covers the spectrum of the IR-G.

The load matrix shown in Fig. 6 gives the linear coefficients for the first three principal components (i.e., PC1, PC2, PC3) formed by the peak intensities of the nine characteristic spectral lines. Figure 6 shows that in terms of PC1, the spectral emission line of Gd corresponds to a higher positive coefficient, and the characteristic spectral intensity of Y corresponds to a higher positive coefficient for PC3. In terms of PC2, the spectral emission lines for Gd and Zn correspond to a higher positive coefficient.

3.3. Phosphor recognition based on supervised ML

3.3.1. Principle of BP-ANN

The BP-ANN supervised ML algorithm was used to identify the phosphor samples. A BP neural network is a type of multilayer feedforward neural network that propagates backward by error. Generally, this type of network adopts a structure composed of an input layer, a hidden layer, and an output layer. The structure of the BP-ANN is shown in Fig. 7. M, I, and J are the numbers of neurons in the input layer, the hidden layer, and the output layer, respectively. X_{m}, K_{i}, and Y_{j} are the input of the mth neuron, the output of the ith neuron in the hidden layer, and the jth neuron in the output layer, respectively, of the BP neural network. The joining weight from X_{m} to K_{i} is ω_{mi}, and the joining weight from K_{i} to Y_{j} is ω_{mi}. The excitation functions of the hidden layer and the output layer are $f(\cdot)$ and $g(\cdot)$, respectively. Here, $f(\cdot)$ is a linear function and $g(\cdot)$ uses the log-sigmoid function, as shown in Eq. (2).

$$g(x) = \frac{1}{1 + e^{-x}} \tag{2}$$

The actual output from BP-ANN is $Y(n)$. The expected output is $d(n)$, where n is the number of iterations. The error signal of the nth iteration is given by $e_j(n)$=$d_{\mathrm{j}}(n)$-$Y_{\mathrm{j}}(n)$. Therefore, the loss function is defined as shown in Eq. (3).

$$e(n) = \frac{1}{2}\sum_{j=1}^{J} e_j^2(n) \tag{3}$$

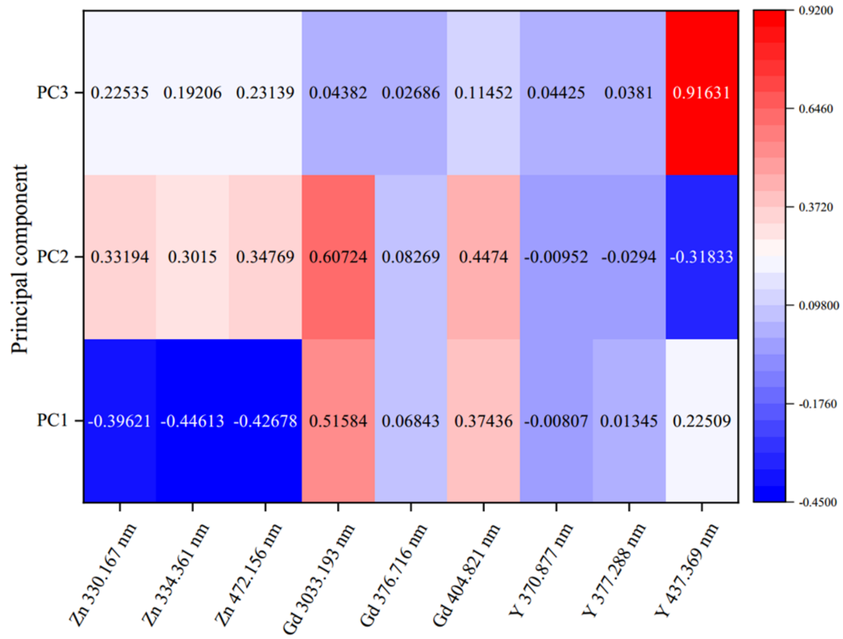

Fig. 6. The relationship between the first 3 PCs and the intensity of 9 spectral lines.

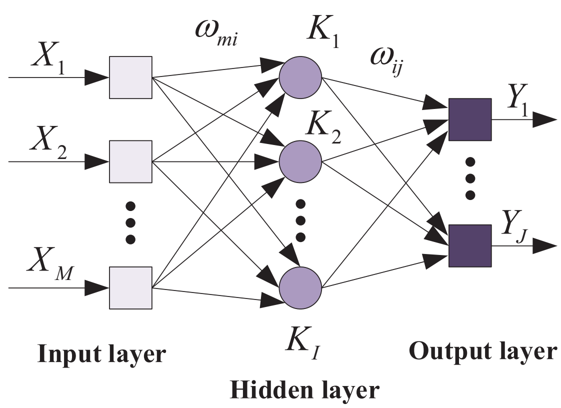

Fig. 7. The structure of BP-ANN.

3.3.2. Improvement of momentum term in BP neural network algorithm

The conventional BP neural network uses the gradient descent method as its main algorithm. The weight ω is adjusted in the direction opposite of the gradient change of the loss function, which means that the error decreases in the gradient direction. After continuous iterative training, the error meets the training accuracy requirement. The parameters are updated according to Eqs. (4) and (5).

$$\omega(n+1) = \omega(n) + \nabla\omega(n) \tag{4}$$

$$\nabla\omega(n) = -\eta * \frac{\partial e(n)}{\partial \omega(n)} \tag{5}$$

In Eqs. (4) and (5), η is the learning rate, and the step length of each iteration. $\nabla\omega(n)$ is the partial derivative of the loss function with respect to the weight ω. Each search is performed

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along the gradient direction of the point, the search path is a zigzag, the convergence speed is slow, the precision is low, and it is easy to fall into local minima. Therefore, Eq. (4) was improved in the work in this paper. The weight updating formula for the improved BP neural network algorithm with an adaptive momentum term is shown in Eqs. (6), (7), (8), and (9).

$$\omega(n+1) = \omega(n) + a(n)\nabla\omega(n-1) + \nabla\omega'(n) \tag{6}$$

$$\nabla\omega'(n) = -\eta * \frac{\partial e(n)}{\partial\omega(n)} \tag{7}$$

$$a(n) = \theta[1 - e^{\rho - MSE(n)}] \tag{8}$$

$$MSE(n) = E[(Y(n) - d(n))^2] \tag{9}$$

In Eq. (6), $\nabla\omega'(n)$ is the partial derivative of the loss function with respect to the weight ω. $a(n)$ is the momentum factor, and θ is a constant used to adjust the speed of at which the curve rises. ρ is a constant and controls the range of change of the loss function. $MSE(n)$ represents the mean square error of the actual output value $Y(n)$ and the expected value $d(n)$. The momentum factor $a(n)$ is adjusted dynamically using $MSE(n)$.

The BP algorithm with the improved momentum term can accelerate the convergence of the algorithm by introducing a last weight update into the current weight update process. With continuous iteration of the algorithm, if the error curve continues to decline, this indicates that the current negative gradient direction is basically the same as the weight change direction from the previous time, and thus the momentum factor $a(n)$ would be increased slowly. If the error curve oscillates, this indicates that the direction of the current negative gradient is in the direction opposite to that of the previous weight change, and thus the momentum factor $a(n)$ would be reduced to mitigate the influence of the previous weight change on the weight update until the error curve did not oscillate.

3.3.3. Establishment of BP neural network model and analysis of the prediction results

Establishment of the BP neural network model involves numerous parameters, including the number of network layers, the number of input layer nodes, the number of hidden layer nodes, the number of output layer nodes, the transfer function, the initial weight and threshold, the learning rate, the maximum number of iterations, the learning objective, the momentum term constant θ, and ρ.

The number of nodes in the input and output layers is determined by the problem to be solved in each case. In this paper, the number of nodes in the input layer depends on the number of characteristic spectra selected when the normalized peak intensity of the spectral lines of the elements in Table 2 is used as the input.

One-hot encoding is used to label the observed values of the different phosphors. Depending on the category of phosphors to which the observed values belong, vectors containing three elements are used to label the different phosphors, as shown in Table 3, where only one element is 1 and the other elements are 0. Therefore, the number of nodes in the output layer is three.

Table 3. Label settings of the observed values

Type of Phosphor	label
CRT-B	001
CRT-G	010
IR-G	100

Because the system is nonlinear, the initial weight has a major influence on whether the learning process can reach the local minimum, whether or not it can converge, and the magnitude of the training time. Therefore, the initial weight is usually selected randomly between [−1, 1]. The learning rate determines the weight change generated in each training cycle. After addition of the momentum term, the learning rate will be adjusted using the momentum term. In this work, the learning rate was set to be 0.001, the maximum number of iterations was set at 5000, the accuracy of learning target was 0.01, the momentum constant θ was 0.1, and ρ was 0.2. The number of nodes in the hidden layer is a key parameter in determining the performance of the BP neural network. Therefore, to determine the number of hidden layer neurons, the original neural network calculation must first be performed. The spectral intensity data samples of the phosphors were divided randomly into two parts, with the training set accounting for 80% and the test set accounting for 20% of the total. After the model was trained and tested, the recognition accuracy was calculated by comparing the predicted tags with the actual tags.

The number of hidden layer neurons was increased gradually from 1 to 10. The recognition rate of the BP-ANN algorithm is shown in Fig. 8. When the numbers of hidden layer neurons are 4, 5, 6, 8, 9, and 10, the recognition accuracy reaches its maximum. Therefore, to improve the classification efficiency while also maintaining high accuracy, the number of hidden layer neurons was set at 4 in this study.

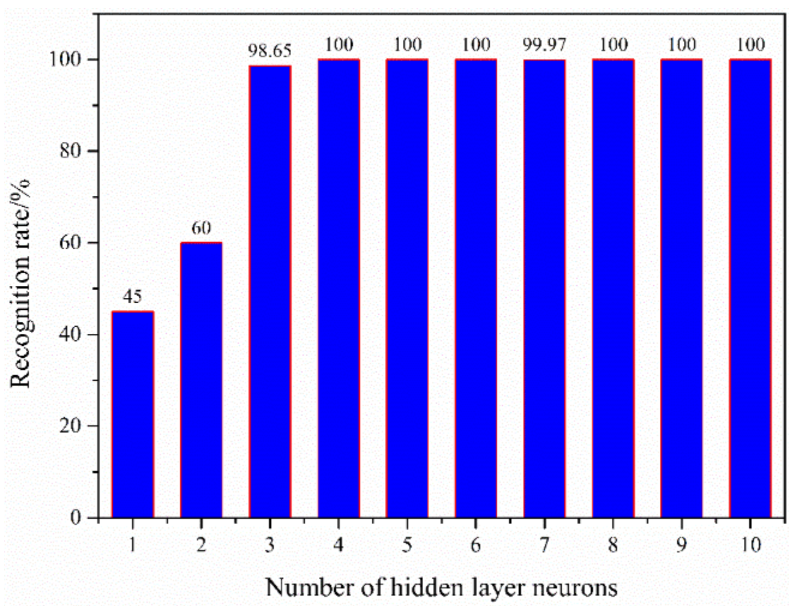

Fig. 8. The recognition rate of BP-ANN with different number of hidden layer neurons.

To provide a better evaluation of the performance of the BP model, the n-fold cross-validation method was used, and n was set at 10. The final recognition accuracy is given by the average of 10 test results. To reduce the differences caused by division of the different samples (data points), the ten-fold cross-validation was repeated ten times, i.e., ten ten-fold cross-validations, and the average of all the results was taken to be the final recognition accuracy.

A classification confusion matrix shown in Fig. 9 is given to exhibit the results of the test data classification run and visualize the prediction accuracy. It is obvious that the prediction accuracy of BP-ANN model is 100% for each phosphor test sample.

As shown in Table 4, when the number of hidden layer neurons is 4, the final recognition accuracy for the phosphor reaches 99.7%, and the recognition rate for each phosphor is also more than 99%. In summary, the classification model composed of four hidden layer neurons

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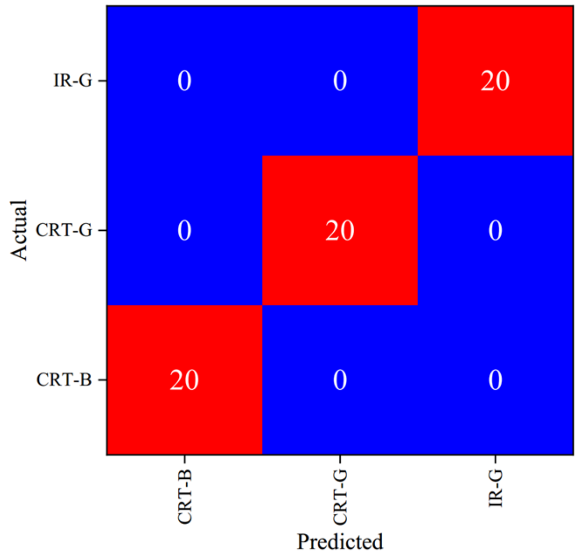

Fig. 9. The classification confusion matrix.

can identify the three different phosphors quickly. Rapid detection of phosphors with LIBS is expected to enable on-line detection and recognition of these phosphors.

Table 4. PCA-BP-ANN parameters and identification results

The number of spectral	The number of principal components	Contribution rate/%	The recognition of different phosphor or total test set/%			
			CRT-B	CRT-G	IR-G	Total test set
9	3	99.7	99.8	100.0	99.8	99.9

4. Conclusion

In this work, a novel system based on a combination of LIBS and ML algorithms is developed, and this self-designed system is applied for identifying and classifying e-waste of REPs. Three different types of phosphors were used as examples. First, the spectrum of the REP IR-G over the wavelength range from 220–470 nm was measured using this new system. The test results showed that there were spectral lines for RE metals such as Gd, Y, and Yb, and for other metals such as Al, Fe, and Ca, after the spectral data were calibrated and analyzed. The spectral lines of Y were numerous and high in intensity, which indicated that the RE element Y is the main constituent element of the IR-G phosphor. Next, we compared the spectra of three different phosphors in the 270–340 nm range. The results show that RE-doped phosphors are quite advantageous because of their high emission efficiency as well as abundant spectral lines compared with metal iondoped phosphors. Furthermore, the spectral data of the phosphors were combined with PCA and the BP-ANN algorithm to classify and identify the three phosphor types (CRT-G, CRT-B, and IR-G). A classification confusion matrix is given to visualize the prediction accuracy. The results show that this new system can successfully complete the classification and prediction of the three different types of phosphors. The recognition rate of the BP algorithm for the test set reaches 99.9%. Ultimately, all results show that this new system combines LIBS with ML algorithms can realize rapid detection and classification of REPs, which will be highly significant for the classification and recycling of RE metals, and for environment protection.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the corresponding author upon reasonable request.

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